# reproducing_np_finite_bug_own_implementation

April 25, 2025

[1]:
```python
import sys
from IPython.display import display, Javascript

def restart_kernel():
    """Restart the Jupyter Notebook kernel to reflect changes in modules and␣
    ↪packages."""
    display(Javascript("Jupyter.notebook.kernel.restart()"))
    print("Kernel is restarting...")

restart_kernel()

import numpy as np
import torch
import tensorly as tl
from tensorly.decomposition import parafac
import sparse
import pickle
import os
import numpy.random as rn
import scipy.stats as st
from tensorly.cp_tensor import CPTensor
import matplotlib.pyplot as plt

from own_implementation import BPTF as BPTF
import bptf

# Please run everything up to the first bptf fit
# Thereafter, run each following cell on their own
```

```
<IPython.core.display.Javascript object>
```

```
Kernel is restarting…
```

# 1 Helper functions

[2]:
```
def generate(shp=(30, 30, 20, 10), K=5, alpha=0.1, beta=0.1):
    """Generate a count tensor from the BPTF model.

    PARAMS:
    shp -- (tuple) shape of the generated count tensor
    K -- (int) number of latent components
    alpha -- (float) shape parameter of gamma prior over factors
    beta -- (float) rate parameter of gamma prior over factors

    RETURNS:
    Mu -- (np.ndarray) true Poisson rates
    Y -- (np.ndarray) generated count tensor
    """
    Theta_DK_M = [rn.gamma(alpha, 1./beta, size=(D, K)) for D in shp]
    Theta_DK_M = [torch.tensor(factor) for factor in Theta_DK_M]
    Mu = tl.cp_to_tensor(CPTensor((None, Theta_DK_M)))
    assert Mu.shape == shp
    Y = rn.poisson(Mu)
    return Mu, Y
```

# 2 Load data

[3]:
```
use_existing_data = False

# for the first bptf.fit:
# using seed 100 causes the assert delta >= 0 to trip
# but using seed 0 doesn't
# please swap between the 2 seeds to get the 2 different types of assertion␣
 ↪errors in the other cells
rn.seed(100)
device = 'cuda'

if use_existing_data:
    assert os.path.exists('sptensor.pkl'), 'No such file.'
    with open('sptensor.pkl', 'rb') as f:
        data = pickle.load(f)
    data = data[:, :, :, :12, :]
    data = torch.tensor(data.todense(), dtype=torch.float64, device=device)
else:
    data = generate(shp=(200, 200, 20, 24, 3), K=10)[1]
    data = torch.tensor(data, dtype=torch.float64, device=device)
```

## 3 Building mask (base example)

[4]:
```
mask = np.ones(data.shape)
# april of GDELT is set to missing
mask[:, :, :, 3, 1] = 0

# diagonals are set to missing
mask[np.eye(mask.shape[0]).astype(bool)] = 0
mask = torch.tensor(mask.astype(np.int64), dtype=torch.float64, device=device)
```

## 4 Fit model

[ ]:
```
n_components = 50
max_iter = 500
tol = 1e-10
verbose = True

BPTF_model = BPTF(data_shape=data.shape, n_components=n_components,␣
 ↪device=device)
BPTF_model.fit(data, mask = mask, max_iter = max_iter, verbose=verbose, tol=tol)
```

```
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When do they occur? []
what is their magnitude? []
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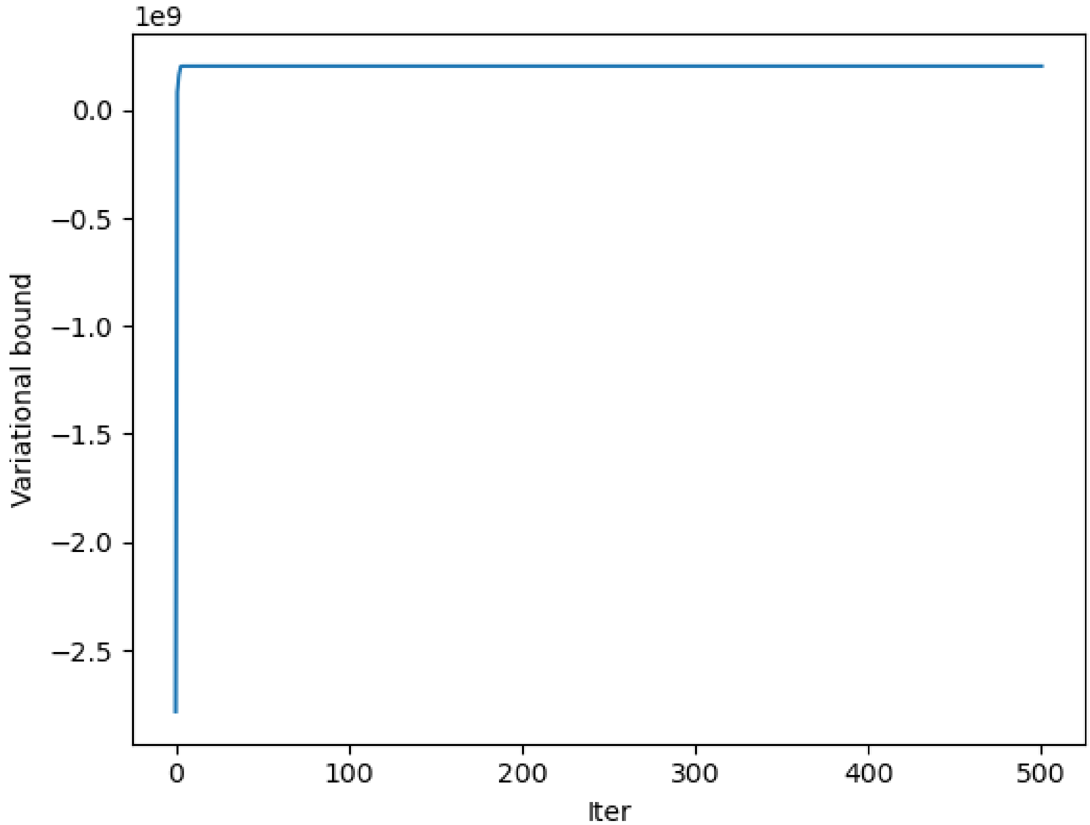


[ ]:
```
BPTF(alpha=tensor([0.1000], device='cuda:0'),
     data_shape=torch.Size([200, 200, 20, 24, 3]), device='cuda',
     n_components=50)
```

## 5 Mask with last mode completely masked

[6]:
```python
mask = np.ones(data.shape)
# april of GDELT is set to missing
mask[:, :, :, 3, :] = 0

# diagonals are set to missing
mask[np.eye(mask.shape[0]).astype(bool)] = 0
```

```python
mask = torch.tensor(mask.astype(np.int64), dtype=torch.float64, device=device)

BPTF_model = BPTF(data_shape=data.shape, n_components=n_components,
  ↪device=device)
BPTF_model.fit(data, mask = mask, max_iter = max_iter, verbose=verbose, tol=tol)
```

```
Number of negative deltas: 0
When do they occur? []
what is their magnitude? []
List of elbos: [-2651132242.5099363, 82082179.42780375, 162077652.1433464,
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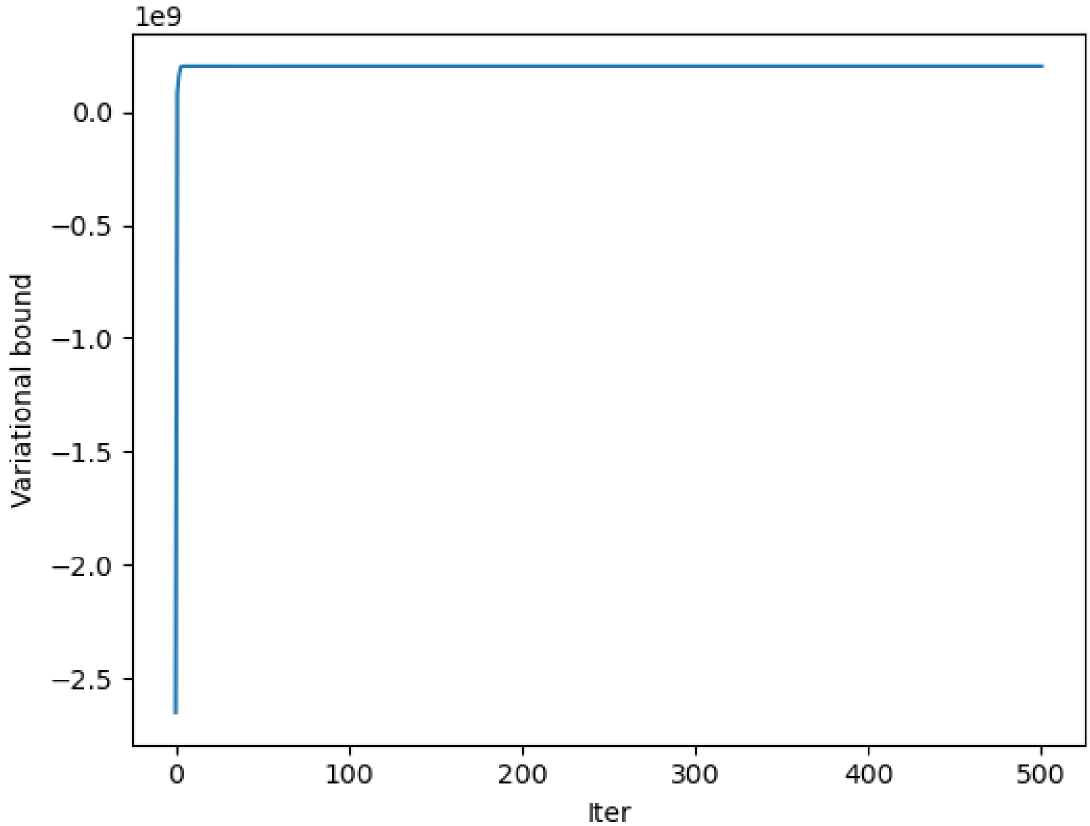


```
[6]: BPTF(alpha=tensor([0.1000], device='cuda:0'),
          data_shape=torch.Size([200, 200, 20, 24, 3]), device='cuda',
          n_components=50)
```

# 6 Mask with the 1st and 3rd indices masked

```
[7]: mask = np.ones(data.shape)
     # april is set to missing
     mask[:, :, :, 3, [0, 2]] = 0

     # diagonals are set to missing
     mask[np.eye(mask.shape[0]).astype(bool)] = 0
     mask = torch.tensor(mask.astype(np.int64), dtype=torch.float64, device=device)

     BPTF_model = BPTF(data_shape=data.shape, n_components=n_components,␣
      ↪device=device)
     BPTF_model.fit(data, mask = mask, max_iter = max_iter, verbose=verbose, tol=tol)
```

```
Number of negative deltas: 0
When do they occur? []
```

```
what is their magnitude? []
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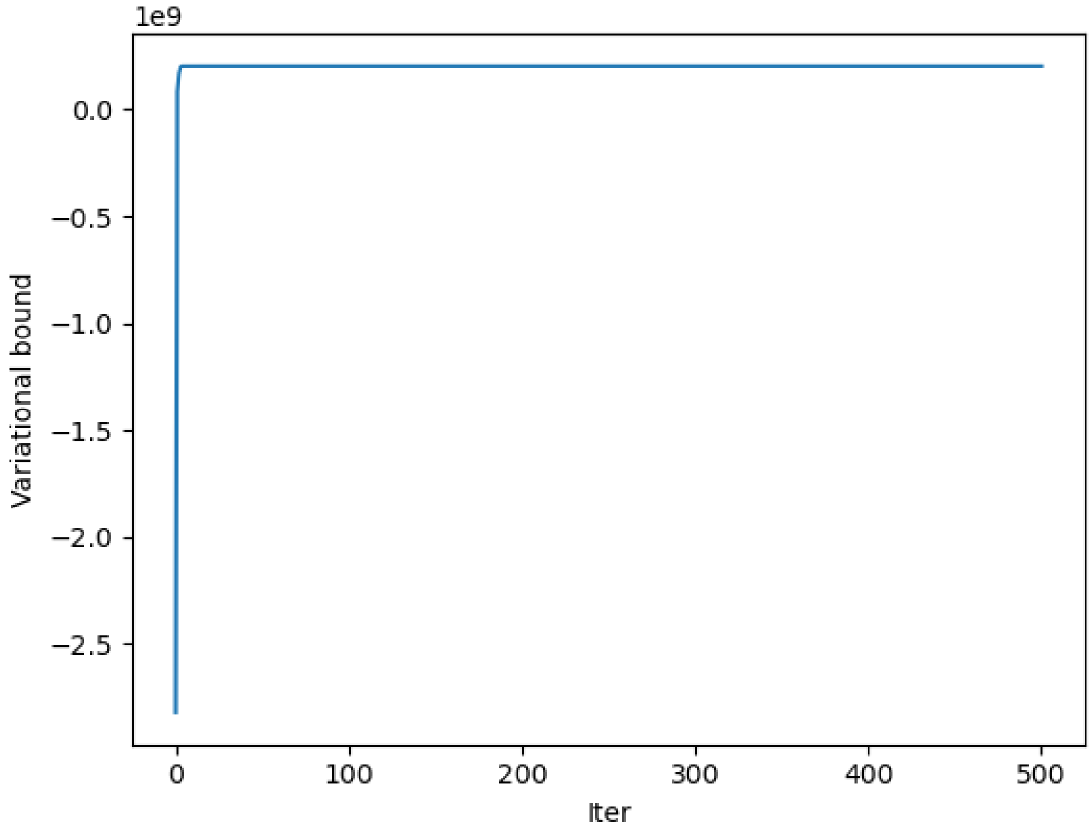


[7]:
```
BPTF(alpha=tensor([0.1000], device='cuda:0'),
     data_shape=torch.Size([200, 200, 20, 24, 3]), device='cuda',
     n_components=50)
```

## 7 Mask with only diagonals masked

[8]:
```
mask = np.ones(data.shape)

# diagonals are set to missing
mask[np.eye(mask.shape[0]).astype(bool)] = 0
mask = torch.tensor(mask.astype(np.int64), dtype=torch.float64, device=device)

BPTF_model = BPTF(data_shape=data.shape, n_components=n_components,␣
 ↪device=device)
BPTF_model.fit(data, mask = mask, max_iter = max_iter, verbose=verbose, tol=tol)
```

```
Number of negative deltas: 0
When do they occur? []
what is their magnitude? []
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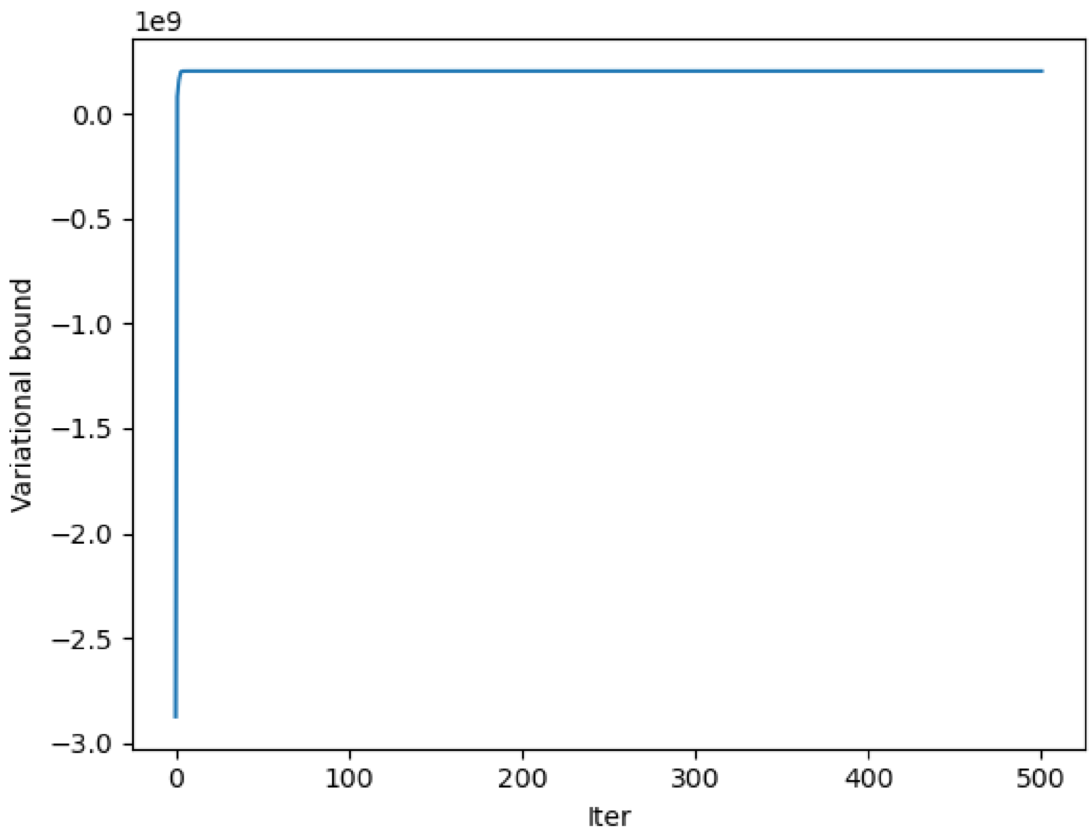


[8]:
```
BPTF(alpha=tensor([0.1000], device='cuda:0'),
     data_shape=torch.Size([200, 200, 20, 24, 3]), device='cuda',
     n_components=50)
```

## 8 No mask

[9]:
```
BPTF_model = BPTF(data_shape=data.shape, n_components=n_components,␣
 ↪device=device)
BPTF_model.fit(data, mask = None, max_iter = max_iter, verbose=verbose, tol=tol)
```

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Number of negative deltas: 0
When do they occur? []
what is their magnitude? []
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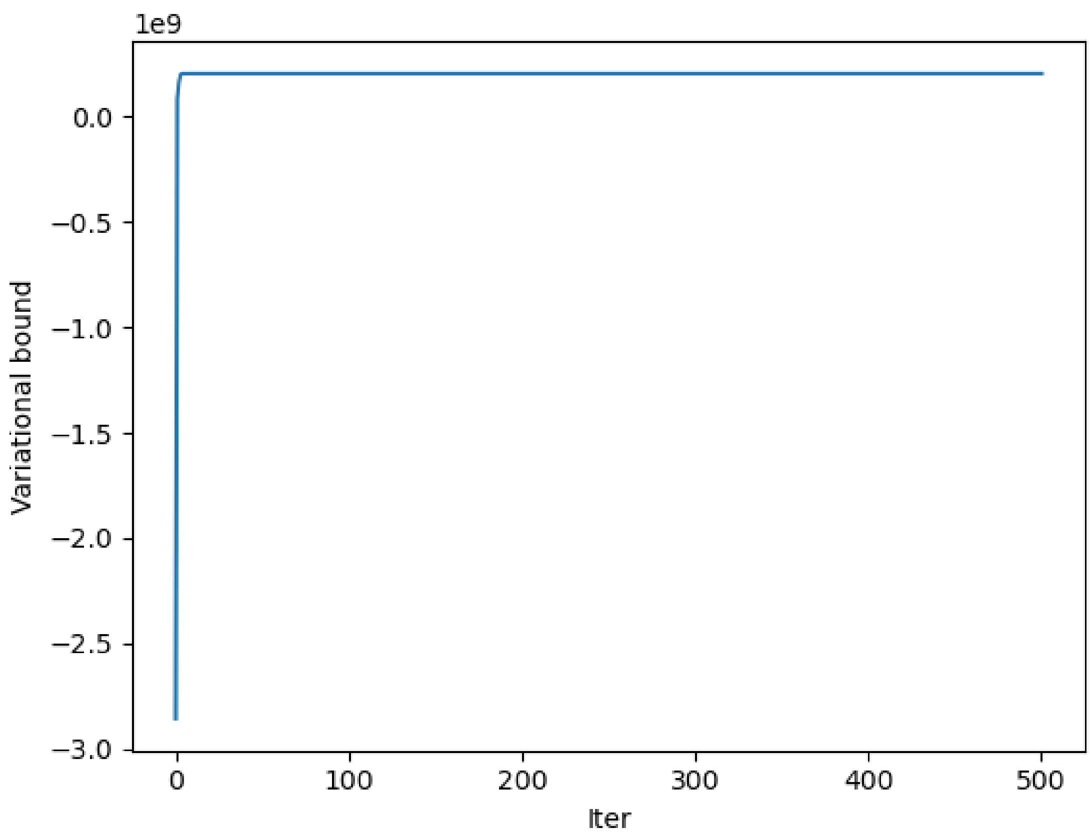

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[9]: BPTF(alpha=tensor([0.1000], device='cuda:0'),
          data_shape=torch.Size([200, 200, 20, 24, 3]), device='cuda',
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